

Lesson Plan #1

Ecohydrology – Grade 11th @ Bowie High School

Relation with my Research: This lesson is linking with my ecohydrology research, which examines how transpiration regulates water movement from plants to the atmosphere under varying environmental conditions. The student experiment reflects the same principles used in scientific studies of plant water use, showing how leaf-level processes scale up to influence the water cycle, agriculture, and regional climate patterns.

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Fellow: Jesus M. Ochoa-Rivero.

Unit: Plant systems and cycling of matter.

Duration: Two Class Periods (90-120 minutes total).

Texas Essential Knowledge and Skills (TEKS) Alignment

Biology TEKS:

B.6 (B) – Analyze the cycling of matter and flow of energy through biological systems.

B.7 (A) – Analyze and evaluate how environmental factors affect photosynthesis and cellular respiration.

B.11 (A) – Analyze the role of internal and external stimuli in maintaining homeostasis.

Texas College and Career Readiness Standards (CCRS):

Interpret and analyze quantitative data.

Construct scientific explanations using evidence.

Develop and interpret graphical representations of data.

Learning Objectives

By the end of this lesson, students will be able to:

1. Define transpiration and explain it using the concept of water potential gradients.
2. Describe the structure and function of stomata in regulating water loss.
3. Design and conduct a controlled experiment to measure transpiration.
4. Calculate and compare rates of transpiration using quantitative data.
5. Construct a scientific explanation using the Claim-Evidence-Reasoning (CER) framework.
6. Explain how transpiration contributes to the water cycle and influences regional climate patterns.

Essential Question

How does transpiration move water from plants into the atmosphere, and how do environmental factors influence its rate?

Key Vocabulary: Transpiration, Water potential, cohesion, adhesion, xylem, stomata, guard cells, evapotranspiration, and homeostasis.

Materials:

- Fresh leaves or small leafy branches
- Plastic bags (Ziploc type)
- String or tape
- Permanent marker
- Stopwatch or timer
- Thermometer
- Digital balance
- Student data table handout

Instructional Procedures
I. Opening Activity (10 minutes)

The teacher/fellow will display an image or short video showing dew formation on leaves or fog rising from a forest.

Students will respond to the following prompts:

- Where does the water on the leaf surface originate?
- How did it reach the leaf?
- Where does it go afterward?

Students will engage in a brief Think-Pair-Share discussion.

The teacher/fellow will state the objective:

“Today we will investigate how plants move water into the atmosphere and measure this biological process using experimental data.”

II. Direct Instruction (15-20 minutes)
Definition and Process

Transpiration is the evaporation of water from plant leaves through stomata. This process is driven by a water potential gradient between the soil, plant tissues, and the atmosphere.

Soil-Plant-Atmosphere Continuum

1. Water is absorbed by plant roots from the soil.
2. Water moves upward through xylem vessels.
3. Evaporation occurs from mesophyll cell walls inside the leaf.
4. Water vapor diffuses out through stomata.
5. Water enters the atmosphere.

Driving Mechanism

Water moves from regions of higher water potential to lower water potential. Evaporation at the leaf surface creates negative pressure (tension), which pulls water upward through the plant via cohesion and adhesion.

Connection to the Water Cycle

Transpiration contributes to evapotranspiration, increasing atmospheric moisture and influencing cloud formation, humidity, and precipitation patterns.

III. Guided Practice – Experimental Design (30-40 minutes)

Research Question

How does _____ affect the rate of transpiration?

(Students select one independent variable: light intensity, temperature, humidity, or leaf surface area.)

Variables

Independent Variable: _____

Dependent Variable: Measured condensation (droplet count, percentage of bag surface covered, or mass loss)

Controlled Variables:

- Type of leaf
- Size of plastic bag
- Observation intervals
- Measurement method
- Time of day

Procedure

1. Place a healthy leaf or small branch inside a clear plastic bag.
2. Seal the bag securely around the stem without crushing the leaf.
3. Place the setup under the assigned environmental condition.
4. Record data at the following intervals:
 - 0 minutes
 - 15 minutes
 - 30 minutes
 - 60 minutes

IV. Data Collection

Students will measure transpiration by one of the following methods:

- Counting visible water droplets
- Estimating percentage of condensation coverage
- Measuring change in mass (if a digital balance is available)

Students must also record environmental conditions such as temperature and light exposure.

V. Mathematical Analysis

Calculating Rate of Transpiration

If using droplet count:

$$\text{Rate} = \text{Number of droplets} \div \text{Time (minutes)}$$

If using mass:

$$\text{Rate} = \text{Change in mass (grams)} \div \text{Time (minutes)}$$

Graphing Requirements

Students must:

- Label the x-axis as Time (minutes)
- Label the y-axis with the transpiration measure and units
- Plot both control and experimental conditions
- Draw a best-fit line
- Calculate the slope

The slope of the line represents the rate of transpiration.

VI. Independent Practice (Second Class Period)

Students will:

- Calculate averages if multiple trials are conducted
- Compare experimental results with control data
- Identify trends and patterns
- Complete a CER-based written conclusion
- Identify at least two potential sources of experimental error

Assessment

Formative Assessment

Exit Ticket:

1. Define transpiration using the term “water potential gradient.”
2. Based on your data, which condition produced the highest transpiration rate and why?

Summative Assessment – Laboratory Report

Students will submit a formal lab report including:

- Hypothesis
- Identification of variables
- Data table
- Graph with labeled axes
- Rate calculations
- Claim-Evidence-Reasoning explanation

- Two sources of experimental error and suggested improvements
- Explanation connecting transpiration to the water cycle

Differentiation Strategies

For English Language Learners:

- Provide vocabulary support sheet
- Provide sentence starters for CER responses
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For Advanced Learners:

- Normalize transpiration rate per unit leaf area
- Estimate transpiration per square meter of vegetation
- Analyze implications for drought conditions in Texas

Closure (5–8 minutes)

The teacher will facilitate a class discussion addressing the following question:

How does transpiration at the plant level influence atmospheric moisture and regional climate patterns?

The lesson will conclude by connecting transpiration to water availability and drought conditions in Texas ecosystems.

Homework Assignment

Students will complete a one-page laboratory reflection including:

- Restated hypothesis
- Summary of methods
- Cleaned data table and graph
- Rate calculation
- CER conclusion
- Two improvements to experimental design
- Explanation of how transpiration contributes to regional water cycling

If you would like, I can now provide:

- A grading rubric (4-level performance scale)
- A student-ready lab handout version
- A formal quiz with answer key
- A version formatted in administrative observation template style
- A downloadable Word document layout structure